

Bernard Kwabena Appiah

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🏠 [Bernard Kwabena Appiah](#) | 📍 [Bernard Kwabena Appiah](#) | 📍 Duluth, MN 55805

RESEARCH INTEREST

Survival Analysis, Statistical Modelling, Biostatistics

EDUCATION

- **University of Minnesota Duluth** Duluth, MN
Master of Science in Mathematical Science (Statistics) August 2025 – 2027
 - GPA: 4.00/4.00
- **University of Ghana** Accra, Ghana
Bachelor of Science in Mathematical Science (Mathematics and Statistics) September 2019 – September 2023
 - Grade: 3.64/4.0
 - First Class Honors
- **Ofori Panin Senior High School** Kukurantumi, Ghana
Secondary Education September 2015 – June 2018
 - Pass with Distinction

PROFESSIONAL EXPERIENCE

- **Department of Mathematics and Statistics, University of Minnesota Duluth** Duluth, MN
Graduate Teaching Assistant August 2025 – Present
 - Lead weekly discussion groups of 20–25 students, breaking down complex course concepts and reinforcing lecture material through interactive activities.
 - Offer one-on-one support during office hours, working closely with students to tackle challenging problems and build their confidence in the subject matter.
 - Provide detailed, constructive feedback on student assignments and exams, helping students identify areas for improvement while celebrating their progress.
- **Department of Statistics and Actuarial Science, University of Ghana** Accra, Ghana
Teaching and Research Assistant (Statistical and Actuarial Consulting Unit) September 2023 – August 2024
 - Assisted the professor by clarifying lecture material, leading discussion sections, and holding review sessions of both undergraduate and graduate level courses.
 - Gathered, cleaned, and analyzed data using statistical software to support research objectives and draw meaningful insights.
 - Evaluated assignments, exams, and projects, providing constructive feedback to help students understand their errors.
 - Conducted thorough reviews of academic articles and research papers to inform study design and provide background context.
- **Ghana Baseline Evaluation Survey** Accra, Ghana
Enumerator and Data Analyst January 2024 – February 2024
 - Conducted door-to-door surveys to collect data on the livelihood of respondents in various communities in the Eastern region.
 - Analyzed the survey results using statistical software to help make informed decisions.
 - Maintained detailed records of survey progress, ensuring accurate and complete documentation.

RESEARCH EXPERIENCE

- **Graduate Thesis** May 2025 – Present
Tool: R Studio
Survival Modeling with Oscillatory Baseline Hazards and High-Dimensional Covariates
 - Developing a survival modeling framework combining damped oscillatory hazard functions with sparse Cox regression.

- Capturing time-varying and non-monotonic risk patterns beyond standard Cox models.
- Applying LASSO and threshold-based methods for variable selection in high-dimensional data.
- Conducting simulation studies to evaluate performance under different hazard structures.
- Applying the model to a breast invasive carcinoma dataset.
- Comparing with Cox, Weibull, and log-normal survival models.

- **Academic Project**

December 2025

Tool: R Studio

Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators

- Implemented a multiple linear regression model using socioeconomic data to predict life expectancy; the final model achieved an Adjusted R^2 of 0.98, successfully identifying Adult Mortality and HIV incidence as the most significant drivers.

A Project Proposal to Investigate the Social, Economic and Legal Aspects of the Galamsey Problem in Ghana

- Proposed a comprehensive research initiative examining the multifaceted issues surrounding the Galamsey problem in Ghana, focusing on social, economic, and legal dimensions.
- The project aims to conduct in-depth analyses of environmental degradation, socioeconomic impact on local communities, and the legal frameworks regulating mining operations in Ghana.

- **Undergraduate Thesis**

January 2023 – September 2023

Tools: R Studio, SPSS, Excel

Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Pregnant Women in Ghana

- Conducted a research project aimed at identifying and analyzing the risk factors contributing to malaria incidence among pregnant women in Ghana.
- Utilized machine learning methodologies, including Random Forest, Decision Trees, K-Nearest Neighbour, Naive Bayes, and Neural Networks, to analyze datasets sourced from the 2019 Ghana Malaria Indicator Survey (GMIS).
- The project sought to uncover significant correlations and predictive models to understand the prevalence of malaria in this vulnerable demographic.

SKILLS

- **Statistical Analysis & Modelling:** R, SPSS, STATA
- **Programming & Database Management:** Python, SQL
- **Technical Documentation & Version Control:** $\text{\LaTeX}$, R Markdown, Microsoft Office Suite

HONORS AND AWARDS

- Summer Research Funding (GAGE), University of Minnesota Duluth, 2026
- Fostering Diversity in Biostatistics Workshop Funding, ENAR, 2026
- Winner, Mathematics Society Quiz Competition, University of Ghana, 2023
- Winner, Statistics and Actuarial Science Quiz Competition, University of Ghana, 2023

LEADERSHIP EXPERIENCE

- **University of Ghana Actuarial and Statistics Society**

Academic Committee Member

2020—2022 Academic Year

- Worked and coordinated with other team members to provide extra tuition for students that needed special assistance.
- Participated in decision-making processes.
- Promoted an atmosphere of academic excellence by sharing academic resources and materials.

- **The Mathematics Society, University of Ghana**

Solution Space Member

2020/2022 Academic Year

CONFERENCES & WORKSHOPS

- ENAR Fostering Diversity in Biostatistics Workshop, ENAR

March 16, 2025

PRESENTATIONS

- *Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators*. Class Presentation, University of Minnesota Duluth. December 2025.
- *Application of Machine Learning Models for Malaria Risk Prediction in Pregnant Women*. Departmental Seminar, University of Ghana. November 2025.

PROFESSIONAL MEMBERSHIPS

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| • American Statistical Association (ASA) | <i>October 2025 – Present</i> |
| • American Mathematical Society | <i>October 2025 – Present</i> |
| • Eastern North American Region International Biometric Society | <i>October 2025 – Present</i> |

CERTIFICATIONS

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| • Exploratory Data Analysis in R – DataCamp | <i>October 2025</i> |
| • Introduction to Regression in R – DataCamp | <i>November 2025</i> |
| • Survival Analysis in R — DataCamp | <i>May 2026</i> |

ADDITIONAL INFORMATION

Languages: English (Highly Proficient; Reading and Writing), Twi (Highly Proficient; Reading and Writing)